Memory Based Question Paper - PGEE 2023

Exam Pattern - 100 Questions (50 Apti and 50 Technical) - 2 sections of 90 mins each.
Difficulty Level of Technical Part - Slightly easier than GATE
Scientific Calculator was similar to that of GATE.

These are a few of the questions I remember from PGEE 2023.

1) Characteristic Equation of JK Flip Flop?
2) Question on stacks and queues. Like push, pop, enqueue dequeue few elements, What will be the final order of the elements?
3) A question on time complexity of min heap
4) Minimum number of NAND gates required for implementing an equation
   (Example - Minimum number of NAND gates required to implement the following binary equation $Y = (\overline{A} + \overline{B})(C + D)$)
5) A question from set theory
   (Example - In an examination 43% passed in Math, 52% passed in Physics and 52% passed in Chemistry. Only 8% students passed in all the three. 14% passed in Math and Physics and 21% passed in Math and Chemistry and 20% passed in Physics and Chemistry. Number of students who took the exam is 200. Then How many students passed in Math only?)

6) GATE-CS-2015 (Kind Corrupt question)

   Consider the following two statements.

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S1: If a candidate is known to be corrupt, then he will not be elected.
S2: If a candidate is kind, he will be elected.
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   Which one of the following statements follows from S1 and S2 as per sound inference rules of logic? **(A)** If a person is known to be corrupt, he is kind **(B)** If a person is not known to be corrupt, he is not kind **(C)** If a person is kind, he is not known to be corrupt **(D)** If a person is not kind, he is not known to be corrupt

7) GATE-CS-2000(Page Fault Question)

   Suppose the time to service a page fault is on the average 10 milliseconds, while a memory access takes 1 microsecond. Then a 99.99% hit ratio results in average memory access time of

   A. 1.9999 milliseconds
   B. 1 millisecond
   C. 9.999 microseconds
   D. 1.9999 microseconds

8) GATE-CS-2004

   A Unix-style i-node has 10 direct pointers and one single, one double and one triple indirect pointers. Disk block size is 1 Kbyte, disk block address is 32 bits, and 48-bit integers are used. What is the maximum possible file size ? **(A)** 2^{24} bytes **(B)** 2^{32} bytes **(C)** 2^{34} bytes **(D)** 2^{48} bytes

9) A similar question from COA

Consider a disk drive with the following specifications:

16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, rotation speed 3000 rpm. The disk is operated in cycle stealing mode whereby whenever one 4 byte word is ready it is sent to memory; similarly, for writing, the disk interface reads a 4 byte word from the memory in each DMA cycle. Memory cycle time is 40 nsec. The maximum percentage of time that the CPU gets blocked during DMA operation is:

A. 10
B. 25
C. 40
D. 50

10) A question to select the subnet mask from given options
11) A similar question from CN

You need to subnet a network that has 5 subnets, each with at least 16 hosts. Which classful subnet mask would you use?

(A) 255.255.255.192

(B) 255.255.255.224

(C) 255.255.255.240

(D) 255.255.255.248

12) A question on Topological Order of the tree
13) Identity element of NOR
14) Number of different bits in the BCD and binary representation of a number
15) A question on BCNF
16) Given functional dependencies -> 3 questions were asked on that, like which NF, number of super keys…
17) Output Based question involving static variable(2-3 output based questions were there)
18) Question related to time complexity of AVL Tree
19) A question on ethernet
20) Minimum number of flip-flops required to construct a mod-50 counter
21) A question in Prims and Krushkals
22) A question on bottleneck spanning tree
23) A question in page replacement algorithm
24) Given 4 statements identify the correct/incorrect statement
25) MSQ on countable and uncountable sets
26) Sum and carry equations of full adder
27) A question on choosing wrong statement made on wifi and bluetooth
28) A question on choosing wrong statement made on VLAN

For Aptitude the difficulty level of the questions was slightly higher than GATE but easier than CAT. I personally followed Arun Sharma(don't solve hard questions) and https://www.indiabix.com/

Topics to focus more on

- Time and Distance
- Problems on Ages
- Simple Interest
- Compound Interest
- Profit and Loss
- Ratio and Proportion
- Percentage
- Alligation or Mixture
- Permutation and Combination
- Area

Few Tips :

1. As somebody who gave both PGEE 2022 and PGEE 2023, I would say the technical part was quite unpredictable. PGEE 2022 had around 8-9 questions from C Programming and Data Structures but PGEE 2023 had more questions from digital logic.
   So revise all the subjects, **DO NOT** leave any subject.
2. PYQs are a goldmine. There were 4-5 GATE PYQs(even the options were the same).
3. No Negative marking. So, attempt all questions.(Please read the instructions to confirm negative marking is not there).
4. Time Management is very important. Do not waste more than a minute on any question.
5. In Aptitude, the last 8-10 questions were reading comprehension and graph based questions, usually people do not have time to do those, so manage time properly as they are easy to score.
6. Lastly, after GATE people usually stop studying so if you can utilize these 2 months properly, there is a high possibility you would be able to crack the exam.

ALL THE BEST!!!